

Lecture 2

→ Elimination is a way to solve the system of equations by doing operations on the coefficient matrix.

→ It is quite similar to trying to express all the variables in just one variable until we find one var and then just keep substituting it to find the others.

→ $Ax = b$

$$\begin{array}{rclclcl}
 \rightarrow \underline{\text{Eq}} & x & + & 2y & + & z & = & 2 \\
 & 3x & + & 8y & + & z & = & 12 \\
 & & & 4y & + & z & = & 2
 \end{array}$$

$$A = \begin{bmatrix} \boxed{1} & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}$$

$$\downarrow 2^R - 1^R \times 3$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & \boxed{2} & -2 \\ 0 & 4 & 1 \end{bmatrix}$$

$$\downarrow 3^R - 2^R \times 2$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 2 & -2 \\ 0 & 0 & \boxed{5} \end{bmatrix}$$

↓ already zero so skipped

→ In case that a pivot turns to be zero, just exchange rows

Failure of elimination →

① There is a case when the final row's pivot turns out to be zero then the matrix is not invertible.

→ Back - substitution step is to repeat the order of operations on the right hand side

→ Augmented matrix →

$$\left[\begin{array}{ccc|c} a_0 & a_1 & \dots & b_0 \\ b_0 & \dots & & b_1 \\ \vdots & & & \vdots \end{array} \right]$$

→ eg back-substitution on previous matrix →

$$\begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix} \xrightarrow{2^R - 1^R \times 3} \begin{bmatrix} 2 \\ 6 \\ 2 \end{bmatrix} \xrightarrow{3^R - 2^R \times 2} \begin{bmatrix} 2 \\ 6 \\ -10 \end{bmatrix}$$

$$\begin{array}{rcl} 5z & = & -10 \\ 2y - 2z & = & 6 \\ 2y + 4 & = & 6 \end{array} \qquad \begin{array}{rcl} \underline{z} & = & -2 \\ y & = & 1 \end{array}$$

→ Instead of doing this we can also perform an addition step of scalar multiplication to get the coefficients as 1.

→ We can also just do the elimination for upper triangle and just find all of the values.

→ Matrices

$$\cdot \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = 3 \begin{bmatrix} \text{col 1} \end{bmatrix} \dots$$

$$\underline{\text{matrix}} \cdot \underline{\text{column}} = \underline{\text{column}}$$

$$\cdot \begin{bmatrix} 1 & 2 & 7 \end{bmatrix} \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} =$$

$$1 \cdot [\text{row 1}] + 2 \cdot [\text{row 2}] \dots$$

$$\underline{\text{row}} \cdot \underline{\text{Matrix}} = \underline{\text{row}}$$

Now the idea is to basically represent the operations we performed during elimination as matrices.

eg. $2^R - 1^R \times 2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ -2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \left[\begin{array}{c} 1 \\ 0 \\ 1 \end{array} \right] \left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right] \left[\begin{array}{c} 1 \\ 0 \\ 1 \end{array} \right]$$

$$\left[\begin{array}{ccc} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \left[\begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$$

→ We can generate all the intermediate matrices let's say E_{21} , E_{31} , E_{32}

$$E = E_{21} \times E_{31} \times E_{32}$$

$$A \times = b$$

$$U \times = c$$

↓
bottom triangle zero

$$\boxed{\begin{array}{l} U = E \times A \\ C = E \times b \end{array}}$$

→ Parentheses can be moved but not order → Associative property

$$E_{32} \times (E_{31} \times (E_{21} \times A))$$

$$= ((E_{32} \times E_{31}) \times E_{21}) \times A$$

→ We saw an example of substitution matrix which subtracts one row from another.

→ Another such operation is the exchange of rows and the matrix which does that is called permutation matrix.
Eg To exchange 2nd & 3rd row

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

→ To exchange columns instead of rows we multiply on right!

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix}$$

→ Inverse Matrix

Basically a matrix that undoes the step (row op / col op) that we performed.

eg $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

$$A^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

$$A \times A^{-1} = I \rightarrow \text{identity matrix}$$

Summary

- Elimination is the process of solving and finding a solution for a system of linear equations.
- We can perform various operations on our coefficient matrix to convert it into a "zero bottom triangle matrix."
- If no solution exists, the matrix A is not invertible.
- Row operations can be represented as matrices → subtraction ops & exchange ops (permutation matrix).
- Inverse is to reverse the order of ops.

